

Part 1: Capturing HTTP Traffic.

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

http

No.

http

Source

Destination

Protocol

Length

Info

666783200

2a02:cb80:4215:2a5a...

2600:1900:4110:86f::

HTTP

334

HEAD /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/9815ce98517045713c3e75cac2f6163...

854998100

2600:1900:4110:86f::

2a02:cb80:4215:2a5a...

HTTP

695

HTTP/1.1 200 OK

772

10.973135100

2a02:cb80:4215:2a5a...

2600:1900:4110:86f::

HTTP

406

GET /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/9815ce98517045713c3e75cac2f6163...

792

11.173033500

2600:1900:4110:86f::

2a02:cb80:4215:2a5a...

HTTP

642

HTTP/1.1 206 Partial Content

1207

15.952753800

2a02:cb80:4215:2a5a...

2600:1900:4110:86f::

HTTP

409

GET /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/9815ce98517045713c3e75cac2f6163...

1211

16.077687800

2600:1900:4110:86f::

2a02:cb80:4215:2a5a...

HTTP

571

HTTP/1.1 206 Partial Content

1341

18.280824500

2a02:cb80:4215:2a5a...

2600:1900:4110:86f::

HTTP

409

GET /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/9815ce98517045713c3e75cac2f6163...

1347

18.440397600

2600:1900:4110:86f::

2a02:cb80:4215:2a5a...

HTTP

174

HTTP/1.1 206 Partial Content

1352

19.472025800

2a02:cb80:4215:2a5a...

2600:1900:4110:86f::

HTTP

410

GET /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/9815ce98517045713c3e75cac2f6163...

1360

19.660829400

2600:1900:4110:86f::

2a02:cb80:4215:2a5a...

HTTP

820

HTTP/1.1 206 Partial Content

1374

20.625288600

2a02:cb80:4215:2a5a...

2600:1900:4110:86f::

HTTP

411

GET /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/9815ce98517045713c3e75cac2f6163...

1390

20.800070000

2600:1900:4110:86f::

2a02:cb80:4215:2a5a...

HTTP

1172

HTTP/1.1 206 Partial Content

1477

22.963419400

2a02:cb80:4215:2a5a...

2600:1900:4110:86f::

HTTP

411

GET /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/9815ce98517045713c3e75cac2f6163...

1484

23.143099800

2600:1900:4110:86f::

2a02:cb80:4215:2a5a...

HTTP

97

HTTP/1.1 206 Partial Content

2171

43.463230600

192.168.1.129

34.223.124.45

HTTP

539

GET /?utm_source=chatgpt.com HTTP/1.1

2176

43.735892900

34.223.124.45

192.168.1.129

HTTP

1027

HTTP/1.1 200 OK (text/html)

> Frame 726: Packet, 334 bytes on wire (2672 bits), 334 bytes captured (2672 bits) on interface 0

> Ethernet II, Src: LiteonTechno_8a:78:6b (f4:6a:dd:8a:78:6b), Dst: NokiaSolutio_65:bf:4d:00:00:00

> Internet Protocol Version 6, Src: 2a02:cb80:4215:2a5a:2458:f9db:895b:bc4b, Dst: 2600:1900:4110:86f::1

> Transmission Control Protocol, Src Port: 61148, Dst Port: 80, Seq: 1, Ack: 1, Len: 260

> Hypertext Transfer Protocol

0000

40 e1 e4 65 bf 41 f4 6a dd 8a 78 6b 86 dd 60 0a

@..e.A.j..xk..

0010

b6 b9 01 18 06 40 2a 02 cb 80 42 15 2a 5a 24 58

.....@*...B*Z\$X

0020

f9 db 89 5b bc 4b 26 00 19 00 41 10 08 6f 00 00

...[.K&...A.o...

0030

00 00 00 00 00 00 ee dc 00 50 17 ff 19 8f 12 c2

.....P.....

0040

7f 65 50 18 00 ff 85 85 00 00 48 45 41 44 20 2f

eP.....HEAD /

0050

65 64 67 65 64 6c 2f 64 69 66 66 67 65 6e 2d 70

edgedl/d iffgen-p

0060

75 66 66 69 6e 2f 67 67 6b 6b 65 68 67 62 6e 66

uffin/gg kkehgbnf

0070

6a 70 65 67 67 66 70 6c 65 65 61 6b 70 69 64 62

jpeggfpl eeakpidb

0080

6b 69 62 62 6d 6e 2f 39 38 31 35 63 65 39 38 35

kibbmn/9 815ce985

0090

31 37 30 34 35 37 31 33 63 33 65 37 35 63 61 63

17045713 c3e75cac

00a0

32 66 36 31 36 33 34 33 34 32 36 34 62 33 66 30

2f616343 4264b3f0

00b0

33 32 64 33 62 61 65 64 37 64 31 61 36 33 34 36

32d3baed 7d1a6346

00c0

62 61 64 30 64 65 64 20 48 54 54 50 2f 31 2e 31

bad0ded HTTP/1.1

Task 1: Filter TCP packets

[illegible]

No.	Time	Source	Destination	Protocol	Length	Info
719	10.656366400	104.121.7.78	192.168.1.129	TLSv1.3	1454	Server Hello, Change Cipher Spec, Application Data
720	10.656367300	104.121.7.78	192.168.1.129	TCP	1454	443 → 61147 [ACK] Seq=1401 Ack=1797 Win=70016 Len=1400 [TCP PDU reassembled in 739]
721	10.656368100	104.121.7.78	192.168.1.129	TCP	1350	443 → 61147 [PSH, ACK] Seq=2801 Ack=1797 Win=70016 Len=1296 [TCP PDU reassembled in 739]
722	10.656368700	104.121.7.78	192.168.1.129	TCP	1454	443 → 61147 [ACK] Seq=4097 Ack=1797 Win=70016 Len=1400 [TCP PDU reassembled in 739]
723	10.656390000	192.168.1.129	104.121.7.78	TCP	54	61147 → 443 [ACK] Seq=1797 Ack=5497 Win=65280 Len=0
724	10.665887800	2600:1900:4110:86f::	2a02:cb80:4215:2a5a::	TCP	86	80 → 61148 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 SACK_PERM WS=256
725	10.665966500	2a02:cb80:4215:2a5a::	2600:1900:4110:86f::	TCP	74	61148 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
726	10.666783200	2a02:cb80:4215:2a5a::	2600:1900:4110:86f::	HTTP	334	HEAD /edged1/diffgen-puffin/ggkkehgbnfjpeggfp1leakpidbkibmmn/9815ce98517045713c3e75cac2f61bmn/
727	10.673488100	2a02:cb80:4215:2a5a::	2a08:1450:4025:2403::	TCP	86	54410 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
731	10.686049900	50.85.242.216	192.168.1.129	TCP	54	443 → 63221 [FIN, ACK] Seq=1 Ack=2 Win=16384 Len=0
732	10.686084000	192.168.1.129	50.85.242.216	TCP	54	63221 → 443 [ACK] Seq=2 Ack=2 Win=251 Len=0
739	10.756004700	104.121.7.78	192.168.1.129	TLSv1.3	1454	Application Data
740	10.756005600	104.121.7.78	192.168.1.129	TLSv1.3	157	Application Data, Application Data
741	10.756041700	192.168.1.129	104.121.7.78	TCP	54	61147 → 443 [ACK] Seq=1797 Ack=7000 Win=65280 Len=0
742	10.759000500	192.168.1.129	104.121.7.78	TLSv1.3	134	Change Cipher Spec, Application Data
743	10.759973800	192.168.1.129	104.121.7.78	TLSv1.3	146	Application Data

> Frame 726: Packet, 334 bytes on wire (2672 bits), 334 bytes captured (2672 bits) on interface Ethernet II, Src: LiteonTechno_8a:78:6b (f4:6a:dd:8a:78:6b), Dst: NokiaSolution_65:bff:4

> Internet Protocol Version 6, Src: 2a02:cb80:4215:2a5a:2458:f9db:895b:bc4b, Dst: 2600:1

> Transmission Control Protocol, Src Port: 61148, Dst Port: 80, Seq: 1, Ack: 1, Len: 260

> Hypertext Transfer Protocol

```

0000  40 e1 e4 65 bf 41 f4 6a dd 8a 78 6b 86 dd 60 0a    @..e.A.j...xk...
0010  b6 b9 01 18 06 40 2a 02 cb 8d 42 15 2a 5a 24 58    ...@.....B.Z$X
0020  f9 d8 69 5b bc 4b 26 00 19 00 41 10 08 6f 00 00    ...[K&....A-O...
0030  00 00 00 00 00 00 ee dc 00 50 17 ff 19 ff 12 c0    .....P.....
0040  7f 65 50 18 00 ff 85 85 00 00 48 45 41 44 20 2f    ..eP..... HEAD -
0050  65 64 67 65 64 6c 2f 64 69 66 66 67 65 6e 2d 70    edged1/diffgen-p
0060  75 66 66 69 6e 2f 67 67 6b 65 68 67 62 6e 66    uffin/gg kkehgbnf
0070  6a 70 65 67 67 67 69 6c 65 65 6b 70 69 64 62    jpeggfp1 leaakpid
0080  6b 69 62 67 6d 6e 2f 39 38 31 35 63 69 39 38 35    kibmmn/9 815ce985

```

Part 3: Capturing and Analyzing UDP Traffic

No.	Time	Source	Destination	Protocol	Length	Info
32	3.992070100	192.168.1.85	192.168.1.255	UDP	308	47300 → 20002 Len=266


```

> Frame 32: Packet, 308 bytes on wire (2464 bits), 308 bytes captured (2464 bits) on interface 0
> Ethernet II, Src: TP-Link_79:91:06 (98:ba:5f:79:91:06), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 192.168.1.85, Dst: 192.168.1.255
> User Datagram Protocol, Src Port: 47300, Dst Port: 20002
> Data (266 bytes)
0000  ff ff ff ff ff ff 98 ba 5f 79 91 06 08 00 45 00  ...._Y....E-
0010  01 26 00 00 40 00 11 b5 22 c0 a8 01 55 c0 a8  &..@.-"....U-
0020  01 ff b8 c4 4e 22 01 12 d9 b0 01 f0 00 01 00 fa  ....N"....-
0030  11 00 00 00 00 00 45 fd f6 48 81 a3 ce ab df b7  ....E-.H....-
0040  d8 bc 9e a4 86 f6 84 eb 89 ec ce e2 c0 a4 c5 b1  ....-.....-
0050  d0 f2 c8 b3 91 f6 84 eb 9e ee b1 d8 bc 9e a4 86  ....-.....-
0060  ab 9a b8 94 b6 db ba d9 fb c1 e3 da e2 cf 8d cc  ....-.....-
0070  e1 d4 92 bf 88 b1 9c a5 94 b9 89 bf 9d b1 93 fa  ....-.....-
0080  8a a8 92 b0 81 b8 8a a4 95 a3 9b b5 84 aa 92 a7  ....-.....-
0090  85 a9 8b e6 89 ed 88 e4 c6 fc de 8c c9 ff ca fa  ....-.....-
00a0  d8 f4 d6 b9 c9 ac de bf cb a2 cd a3 fc 91 fe 9a  ....-.....-
00b0  ff dd e7 c5 97 d2 f0 dc fe 8e fc 93 f7 82 e1 95  ....-.....-

```

Part 4: Comparing TCP and UDP by filling in the following tables. Save your work (e.g., in an MS Word document), and upload it to your online git repo.

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	Reliable and creates a connection before sending data.
Data Integrity and Ordering	TCP	Keeps data in order and checks for errors.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Websites, Email, File Transfer	Video Streaming, Online Games, Voice Calls
Performance	Slower but reliable	Faster but less reliable